An OpenFOAM Paper Written Without the Template

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Abstract

This is the place for an abstract.

1 Introduction

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1.1 Subsection

Example text:

We shall consider the specific transport property ϕ and note that its spatial and temporal variation is governed by a second-order partial differential equation (PDE), viz.

$$\frac{\partial}{\partial t}(\rho\phi) + \nabla\bullet(\rho\phi\mathbf{U}) - \Gamma_\phi\nabla^2\phi - S_\phi(\phi) = 0. \tag{1}$$

Herein, $\phi = \phi(\mathbf{x}, t)$ is an arbitrary general intensive physical quantity, e.g., a fluid property (scalar or tensor of any rank). Thus, Eqn. 1 is often referred to as a generic transport equation.

OpenFOAM® (Open Field Operation And Manipulation) is a flexible and mature C++ Class Library for Computational Continuum Mechanics (CCM) and Multiphysics. Its Object-Oriented-Programming (OOP) paradigm enables to *mimic data types and basic operations* of CCM using top-level syntax as close as possible to the conventional mathematical notation *for tensors and partial differential equations*, as shown in Lst. 1.

Listing 1: code listing caption

```
solve
(
  fvm::ddt(rho,Phi)
  + fvm::div(phi,  Phi)
  - fvm::laplacian(Gamma,  Phi)
```

Table 1: Finite Volume Notation

implicit differential operators	
rate of change	$\mathrm{J}\frac{\partial[\rho\phi]}{\partial t}\mathrm{K}$
convection term	$\mathrm{J}\nabla\bullet\left(F[\phi]_{f(F,S,\gamma)}\right)\mathrm{K}$
diffusion term	$\mathrm{J}\nabla\bullet(\Gamma\nabla[\phi])\mathrm{K}$
linear part of source term	$\mathrm{J}S_p\,[\phi]\mathrm{K}$
explicit differential operators	
temporal term	$\frac{\partial\rho\phi}{\partial t}$
divergence term	$\nabla\bullet(\rho\mathbf{U}\phi_{f(\rho\mathbf{U},S,\gamma)})$
laplacian term	$\nabla\bullet(\Gamma\nabla\phi)$
constant part of source term	S_u

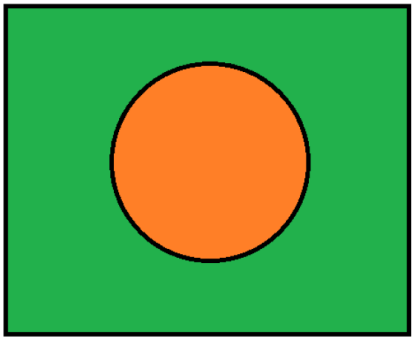


Figure 1: Examplary figure

```
  ==
    Sphi
);
```

Besides providing OpenFOAM® code itself, spatial and temporal discretisation of Eqn. 1 can also be described in a precise and concise manner using the finite-volume notation [**?**] - see Tab. 1.

2 Theoretical backgroud

Text in this section. Here is an example figure (Fig. 1).

3 Conclusion

This is a conclusion. Notice how equations, tables and figures are referred to in the text as Eqn. 1, Tab. 1 and Fig. 1, respectively.